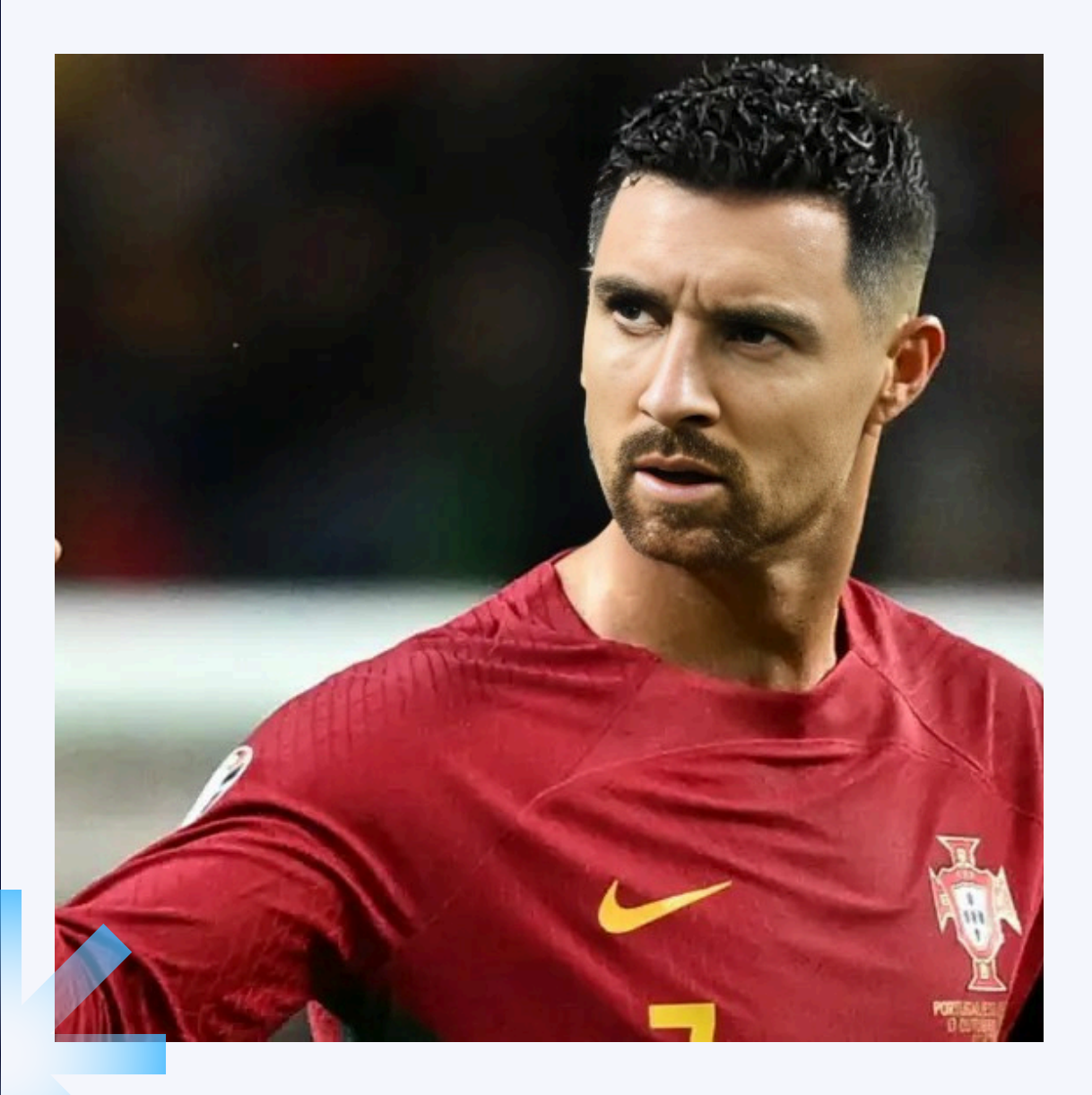


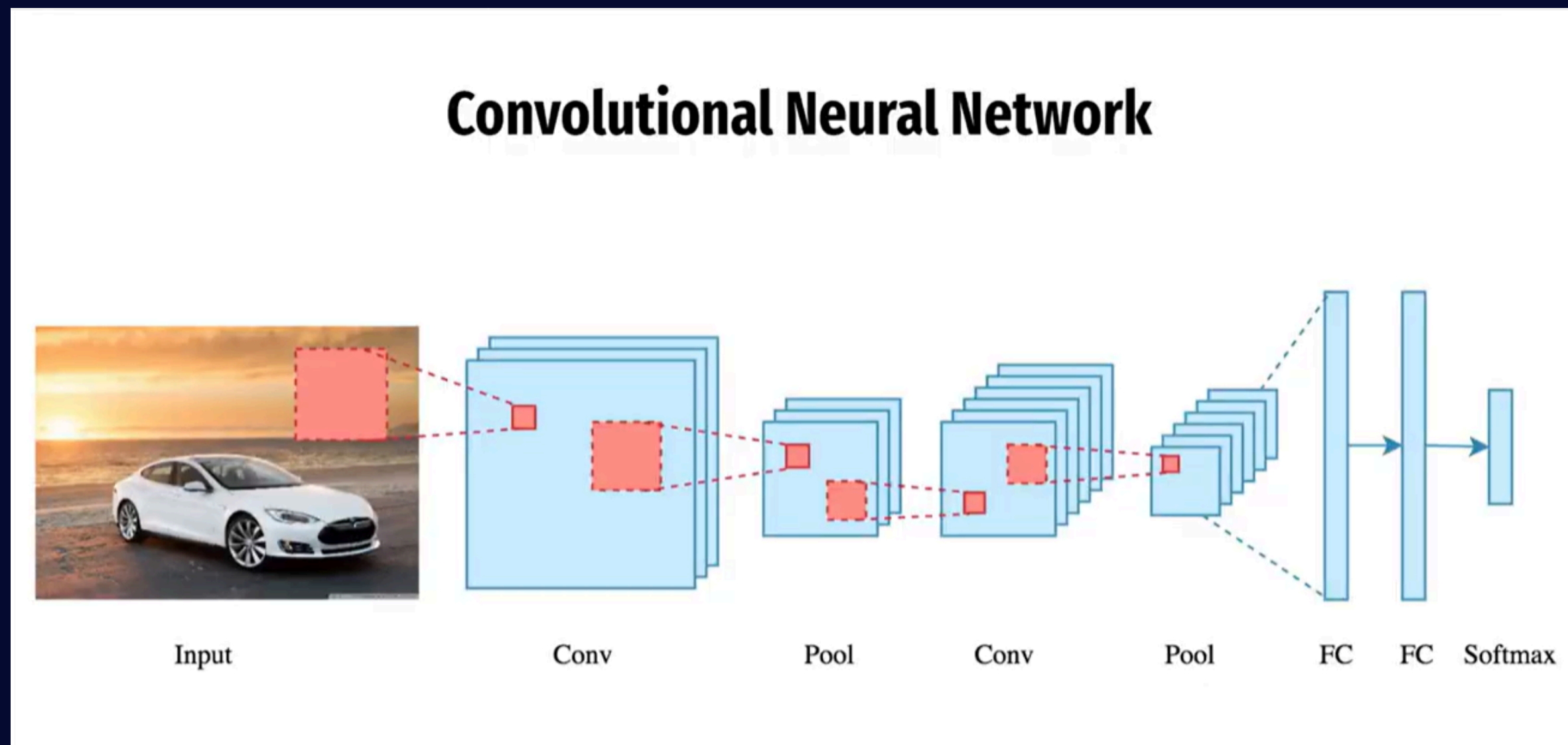


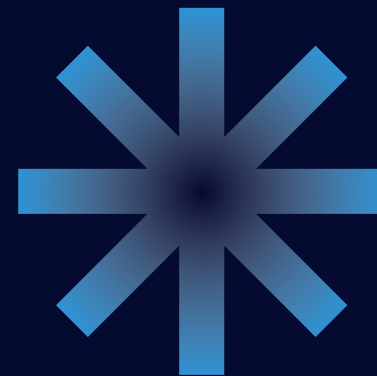
DEEPPFAKE DETECTION

Presented by Group 5

CNN

A Convolutional Neural Network (CNN), also known as ConvNet, is a specialized type of deep learning algorithm mainly designed for tasks that necessitate object recognition, including image classification, detection, and segmentation. CNNs are employed in a variety of practical scenarios, such as autonomous vehicles, security camera systems, and others.





Input image

9	4	1	2	2
1	1	1	0	4
1	2	1	0	6
1	0	0	2	1
9	6	7	4	1

Filter

0	2	1
4	1	0
1	0	1

Output array

Output [0][0] = $(9*0) + (4*2) + (1*4)$
+ $(1*1) + (1*0) + (1*1) + (2*0) + (1*1)$
= $0 + 8 + 1 + 4 + 1 + 0 + 1 + 0 + 1$
= 16

16		

CONVOLUTIONAL LAYER

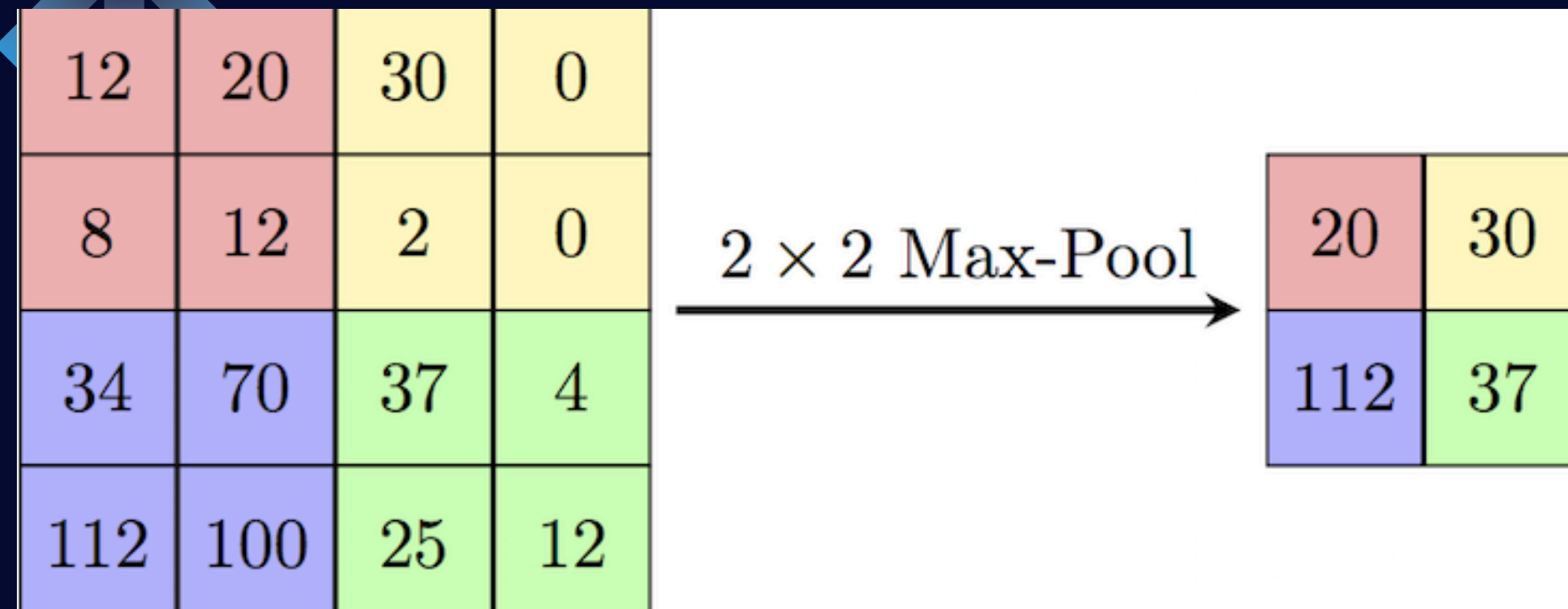
The convolutional layer is the core building block of a CNN, and it is where the majority of computation occurs.

There are hyperparameters which affect the volume size of the output that need to be set before the training of the neural network begins:

- The number of filters
- Stride
- Padding

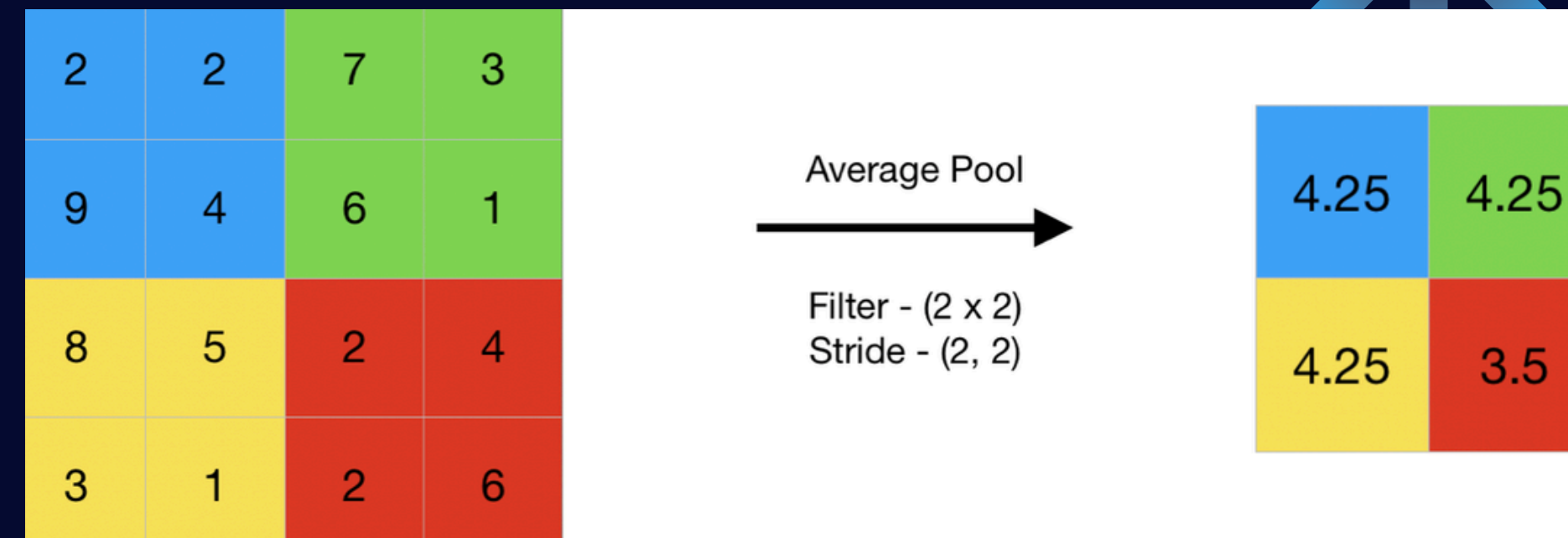
POOLING LAYER

Pooling layers, also known as downsampling, conducts dimensionality reduction, reducing the number of parameters in the input. While a lot of information is lost in the pooling layer, it also has a number of benefits to the CNN. They help to reduce complexity, improve efficiency, and limit risk of overfitting.



Max pooling

As the filter moves across the input, it selects the pixel with the maximum value to send to the output array. As an aside, this approach tends to be used more often compared to average pooling.

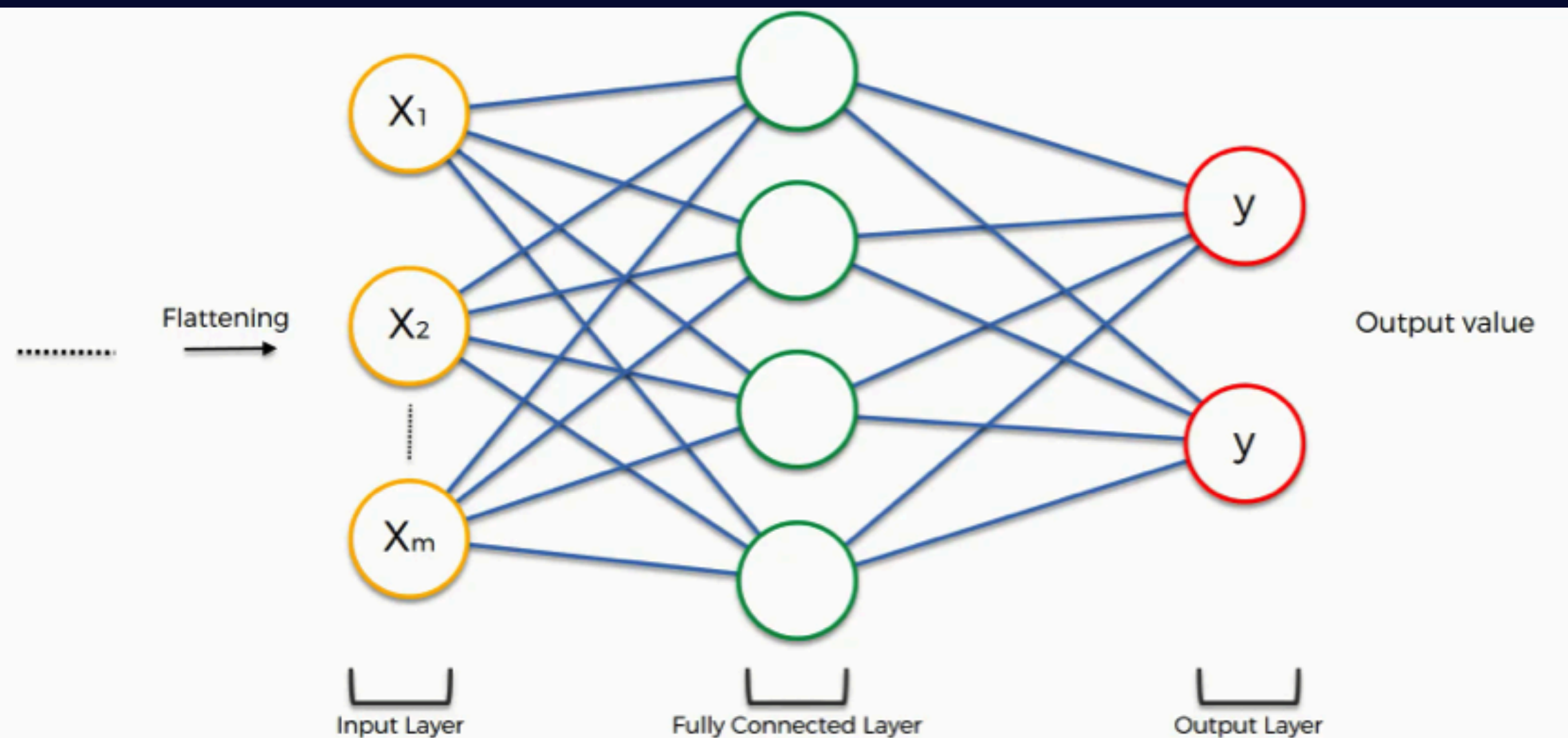


Average pooling

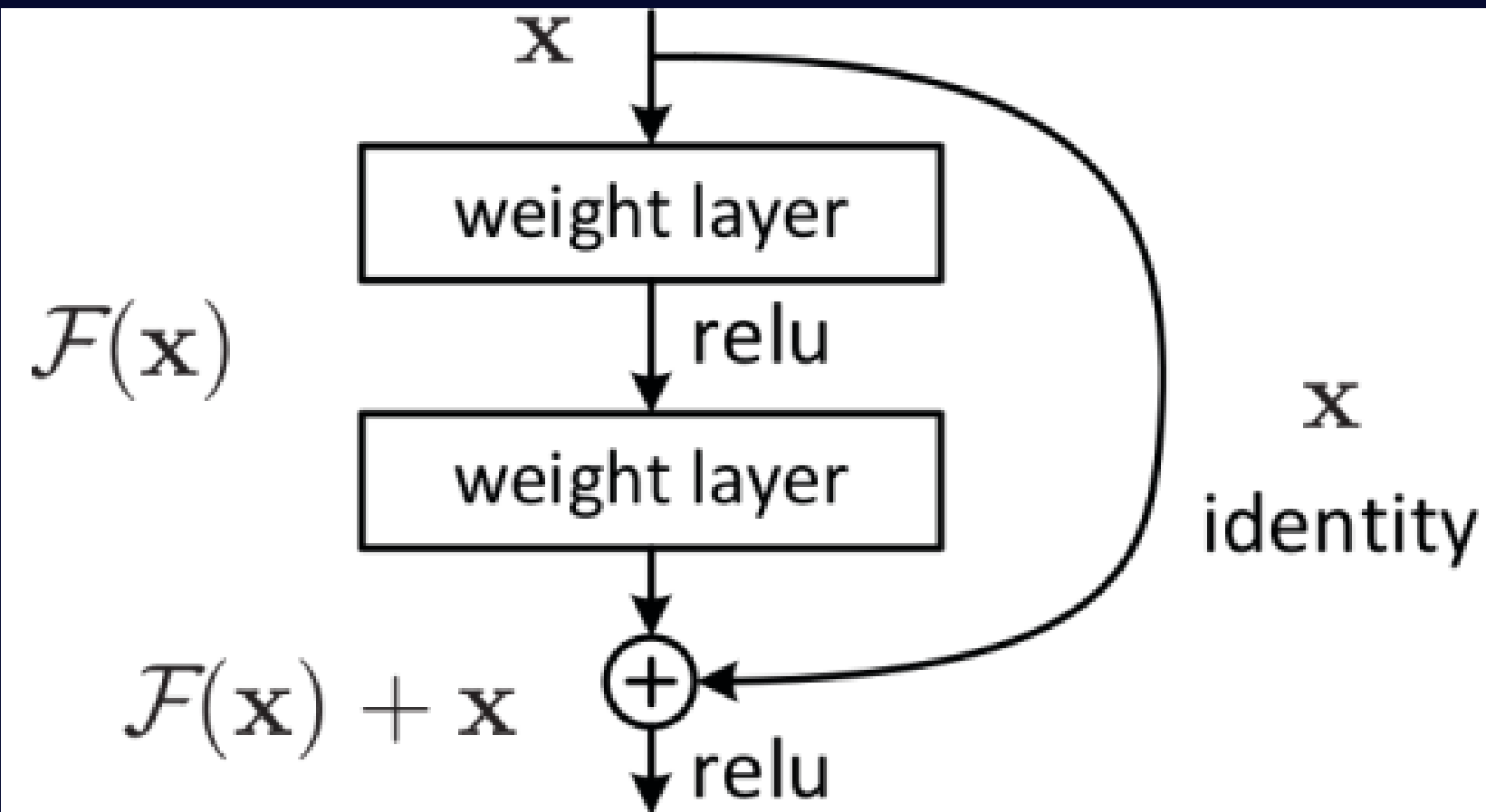
As the filter moves across the input, it calculates the average value within the receptive field to send to the output array.

FULLY-CONNECTED LAYER

In a fully connected layer, every neuron in the layer is connected to every neuron in the previous layer. This means each output neuron computes a weighted sum of all inputs, adds a bias, and passes the result through an activation function.

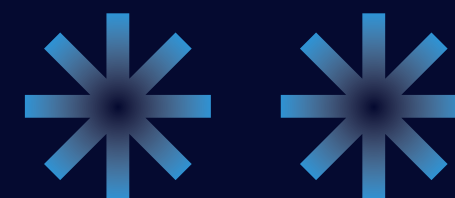


RESNET



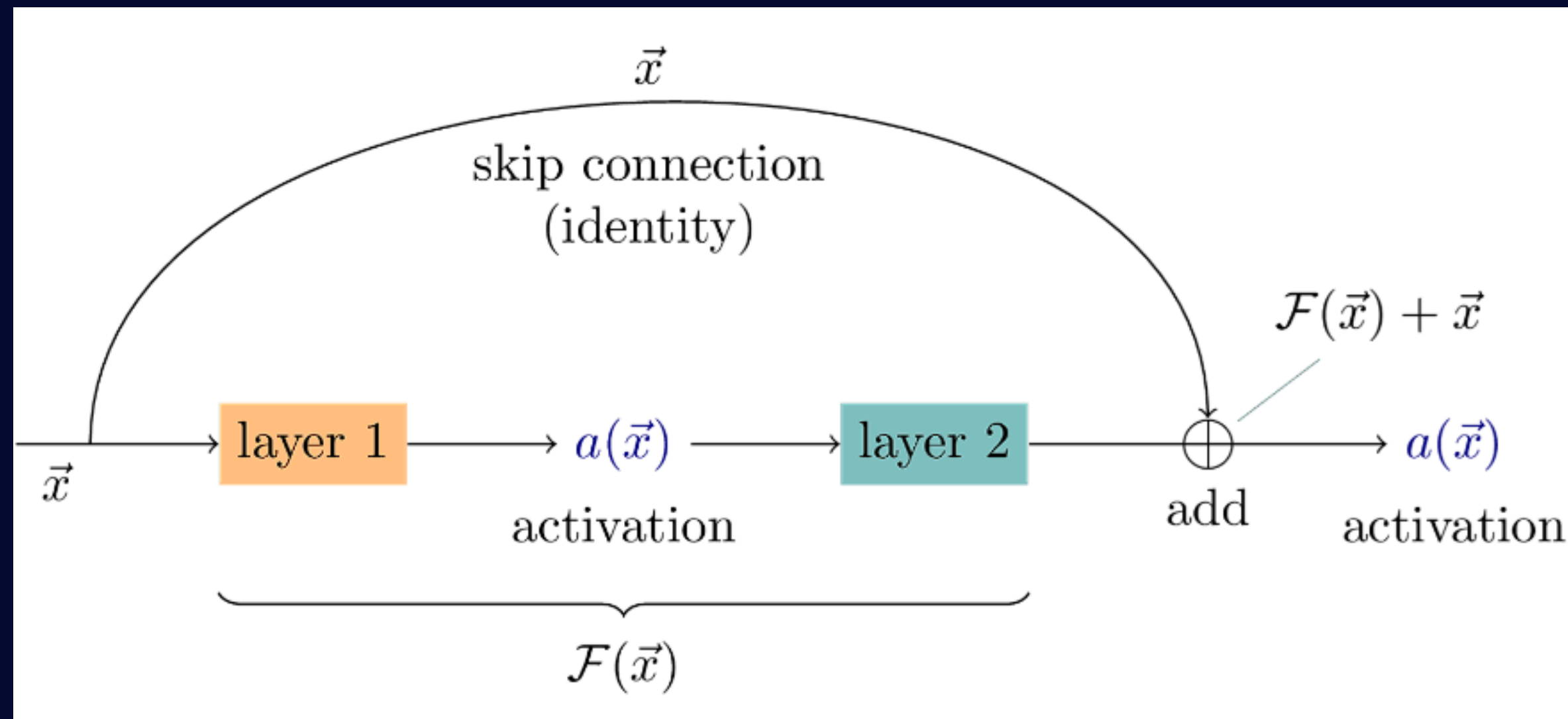
ResNet (Residual Network) is a popular deep convolutional neural network architecture introduced by Microsoft Research in the paper "Deep Residual Learning for Image Recognition"

ResNet is a CNN designed to work with hundreds of layers. It uses skip connections (residual connections) to address the vanishing gradient and degradation problems.



SKIP CONNECTION

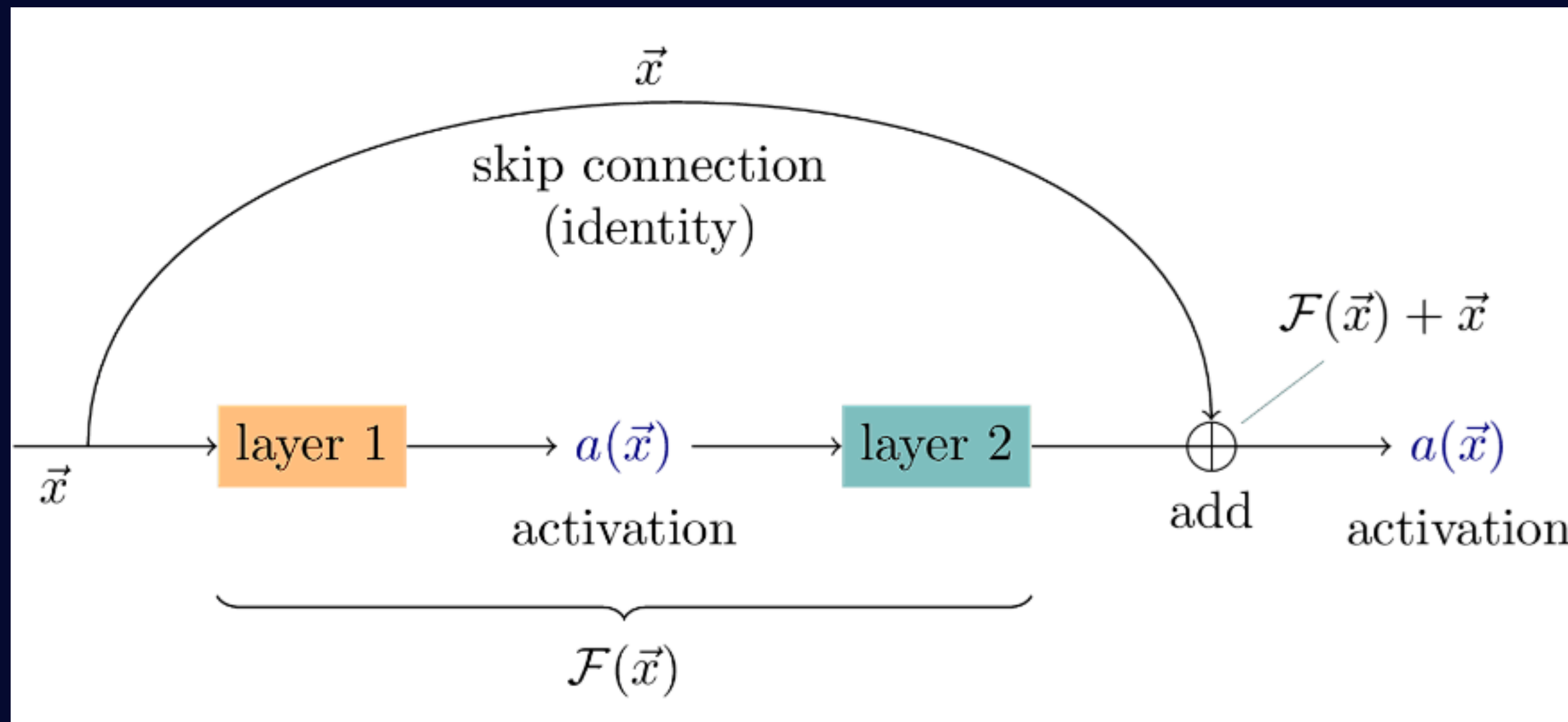
Skip Connections (or Residual Connections) as the name suggests skips some of the layers in the neural network and feeds the output of one layer as the input to the next layers



SKIP CONNECTION

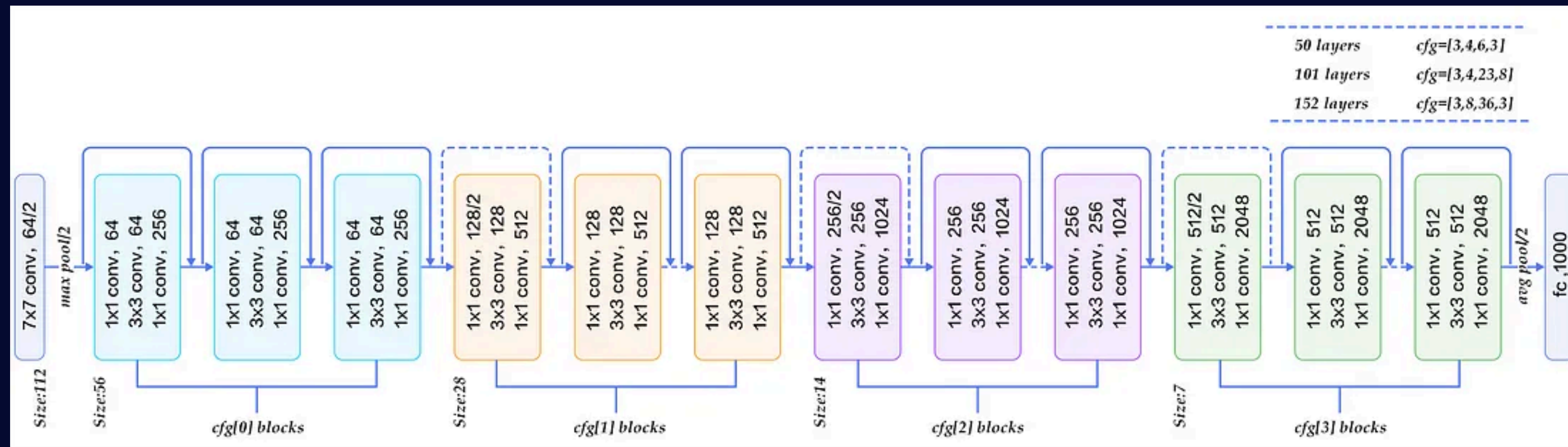
$$H(x) = F(x) + x$$

- x : Input of the residual block
- $F(x)$: output after x passes through a sequence of layers: Conv $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Conv $\rightarrow$ BN



RESNET50

ResNet-50 is a specific version of the ResNet (Residual Network) architecture with 50 deep layers. It is one of the most popular architectures in computer vision due to its high performance and strong generalization ability across various tasks such as image classification, object detection, and face recognition.



DEEP FAKE



- Tạo ra hình ảnh, video của người khác dù không có đặc điểm của họ
- Các thuật toán deep fake ngày càng tinh vi, phức tạp, khó để phát hiện hơn



DEEP FAKE

- Lộ rõ rủi ro tiềm ẩn của deep fake vào mục đích xấu:
 1. Làm video, hình ảnh giả mạo để lừa đảo
 2. Xúc phạm, hạ uy tín của người bị đánh cắp khuôn mặt
 3. Giả mạo lời nói, hành động của lãnh đạo, gây hoang mang





THANK YOU